

SatQuery-Edge

Offline Satellite Intelligence Investigation Report
SIH 26167 · ISRO · Prototype · Offline Mode

Investigation ID	SQE-TEST-001
Timestamp	2026-09-02 19:18:03
Intent	bi_temporal_change
Tools Used	change, grounding, geospatial

Query

Test query

Final Answer

Test answer

Evidence Score

Component	Code	Value	Weight
Model Agreement	M	0.876	0.30
Sensor Reliability	S	0.900	0.20
Spatial Consistency	P	0.947	0.20
Temporal Consistency	T	0.940	0.15
Input Quality	I	0.600	0.15
FINAL SCORE		86.3 / 100	

Validation

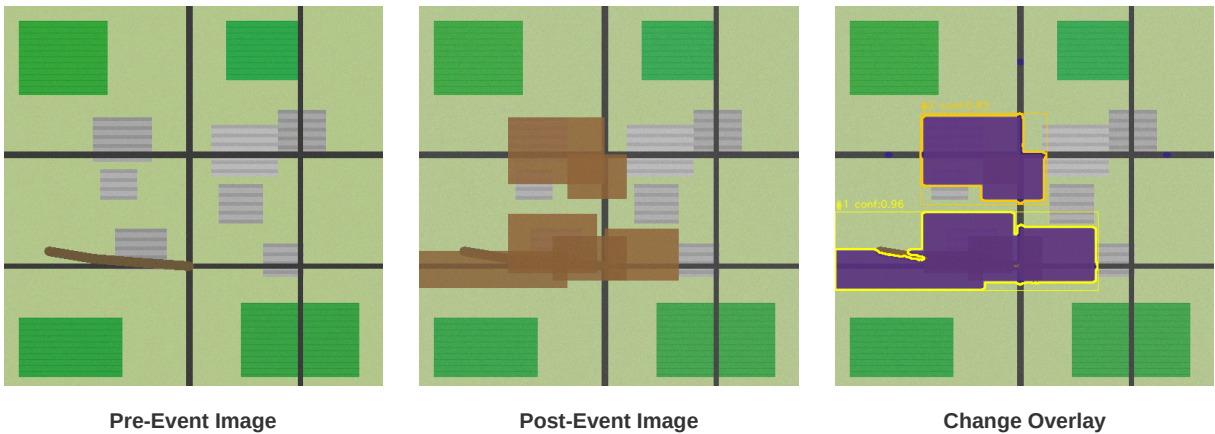
Status: **PASS** ✓

- 2 image(s) detected — count OK.
- Image 1: Quality OK (512×512, 3 channel(s), mean=164, std=37.9).
- Image 2: Quality OK (512×512, 3 channel(s), mean=160, std=36.3).
- Image pair: identical dimensions — perfect pixel grid alignment.

Detected Regions & Approximate Coordinates

#	Label	Confidence	Area (km²)	Demo Georeference
1	candidate changed region	0.96	18.215	Yes
2	candidate changed region	0.83	9.770	Yes

Key Visual Evidence



Specialist Tool Outputs

CHANGE — SUCCESS — 2 region(s)

Detected 2 significant candidate changed region(s).

Largest region:

Approximate change area: 10.3% of scene.

Spectral Change Confidence: 0.96

Center: (0.37, 0.66) (normalized pixel space)

Total changed pixels: ~16.8% of observation area.

Remote Sensing Analysis (NDWI / Inundation Dynamics):

Candidate newly inundated / flooded regions where spectral reflectivity changed significantly b

GROUNDING — SUCCESS — 0 region(s)

Grounding DINO + SAM found no candidate regions matching 'water'.

GEOSPATIAL — SUCCESS — 0 region(s)

Geospatial conversion complete.

Converted 2 region(s) to geographic coordinates.

Total approximate area: 27.98 km².

■ Location Georeferenced to: Bhubaneswar, Odisha (Default Demo)

Center Coords: (20.2960°N, 85.8240°E)

Approximate pixel-to-geographic affine scale applied.

Execution Trace

Tool Versions

Component	Version
SatQuery-Edge Prototype	1.0.0-SIH26167
Python OpenCV	4.11.0
NumPy	1.26.4
Streamlit	1.62.0
Report Engine	ReportLab PDF

Mode	OFFLINE — No external APIs
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NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. —
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